

towards MAPPO and Inverse DRL for MATSim Replanning

A MAPPO extension that learns person-level replanning while retaining MATSim's QSim and scoring engine

PPO with imitation warm-up

Count-aware feedback

Dimensionless state encoding

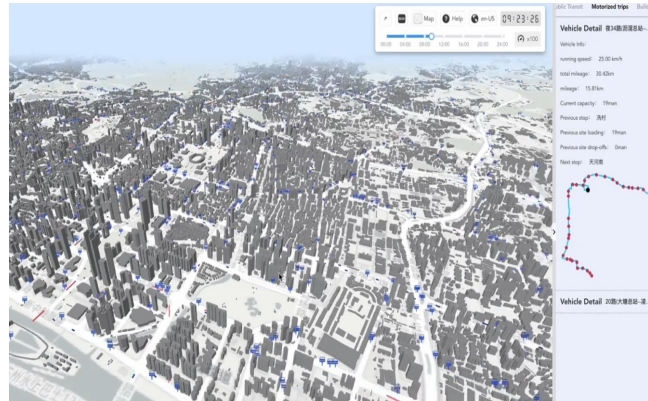
Ming Lü
09.2026

Why MATSim-mappo (Multi-Agent Proximal Policy Optimization)

What if deep reinforcement learning is applied to activity chains?



**Trajectory completion
in surveillance blind spots**

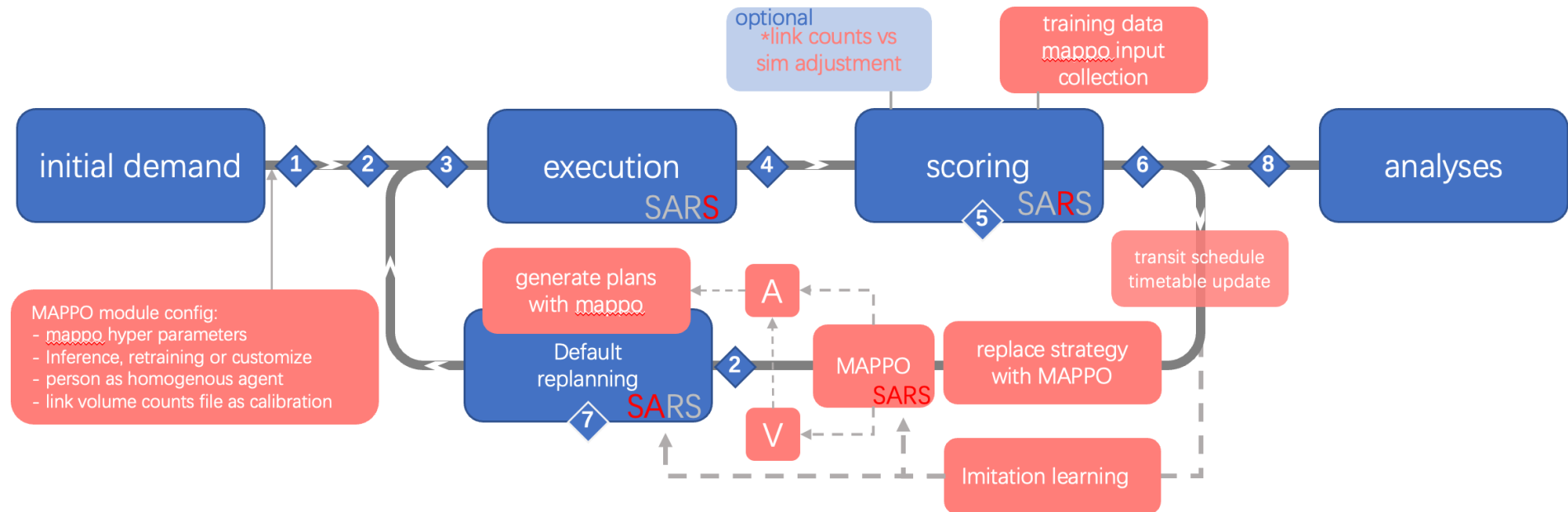


**Public transit optimization from
both demand and supply side**



**UAM scenarios fully utilize
facilities and dynamic avoidance**

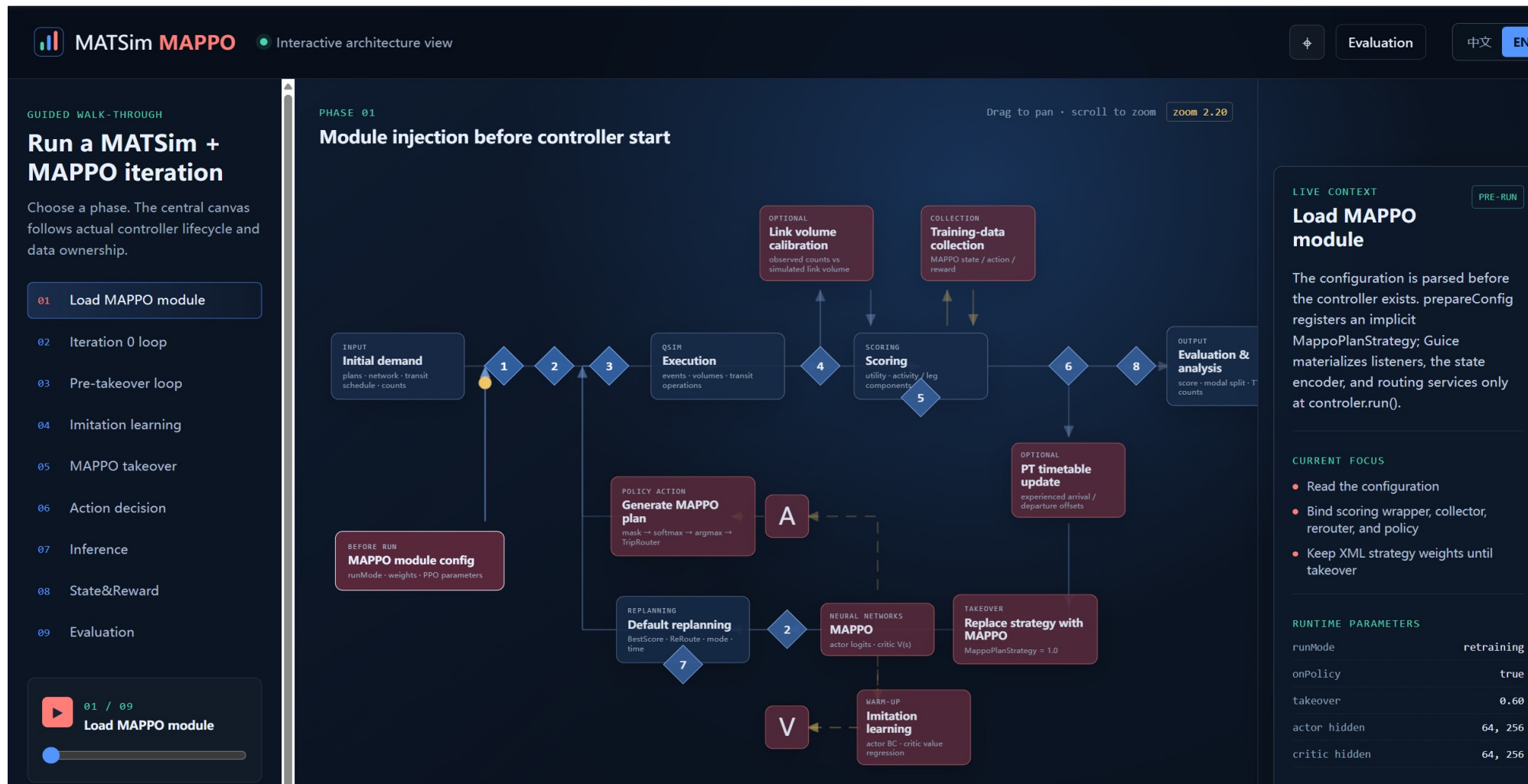
Framework of MATSim-mappo



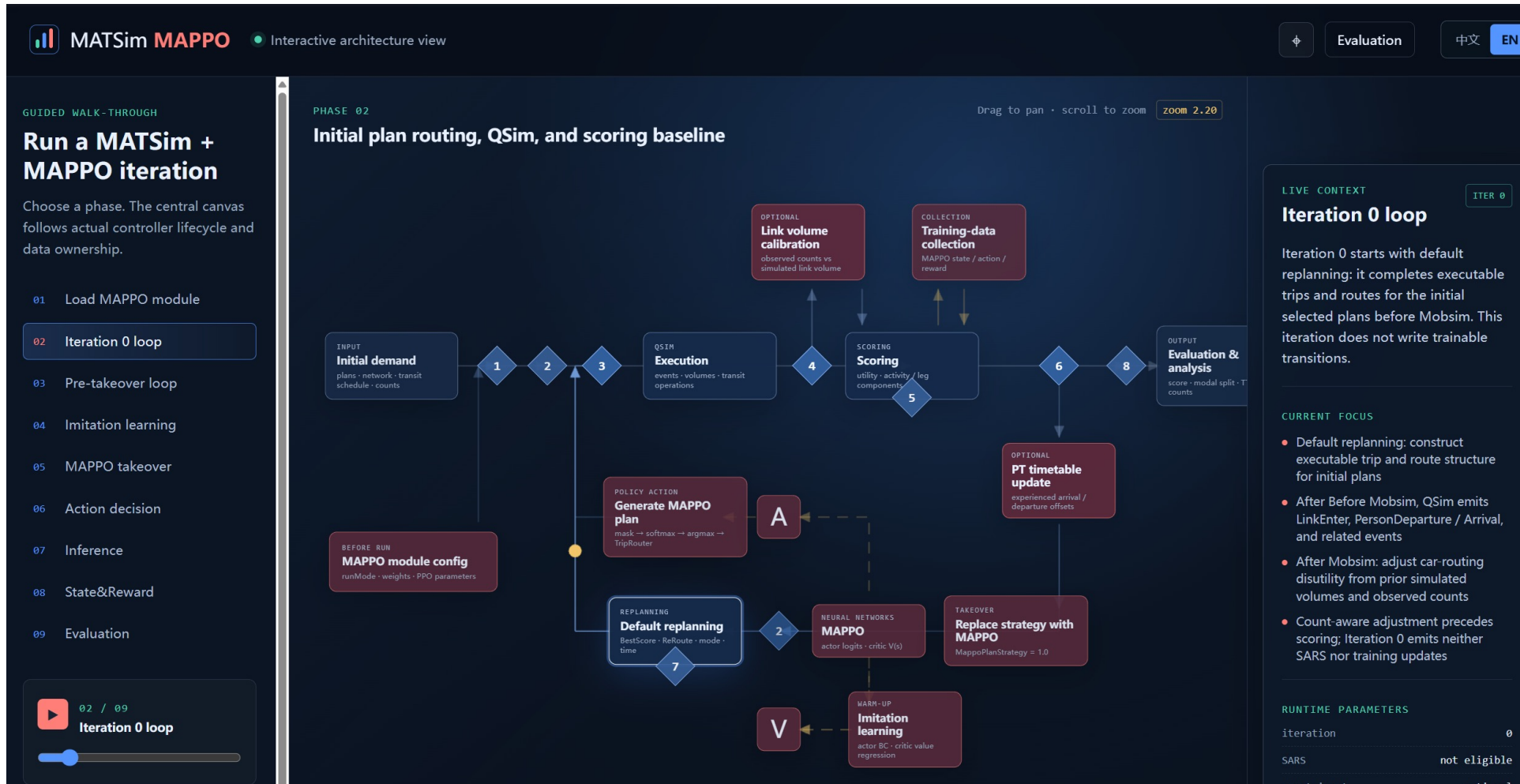
Before takeover: default strategies provide action labels. After takeover: MAPPO generates the candidate plan and receives scored trajectories.

Mappo procedure

Can be visualized at [~ /matsimMappo/matsimMappo/visualization/index.html](https://matsimMappo/matsimMappo/visualization/index.html)



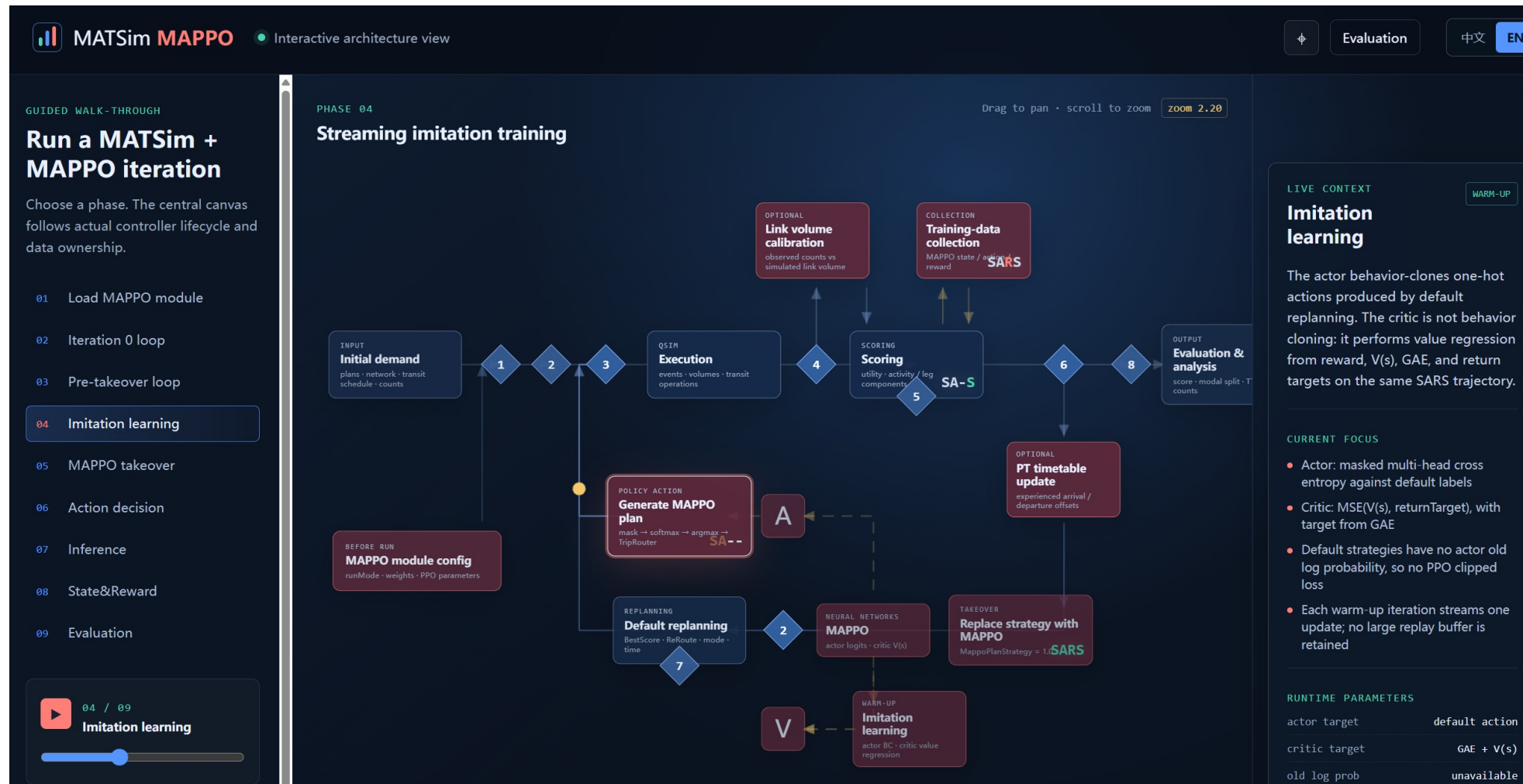
Mappo procedure – running default iterations iter 0-n



Mappo procedure – iterations before mappo strategy takes over iter n-1



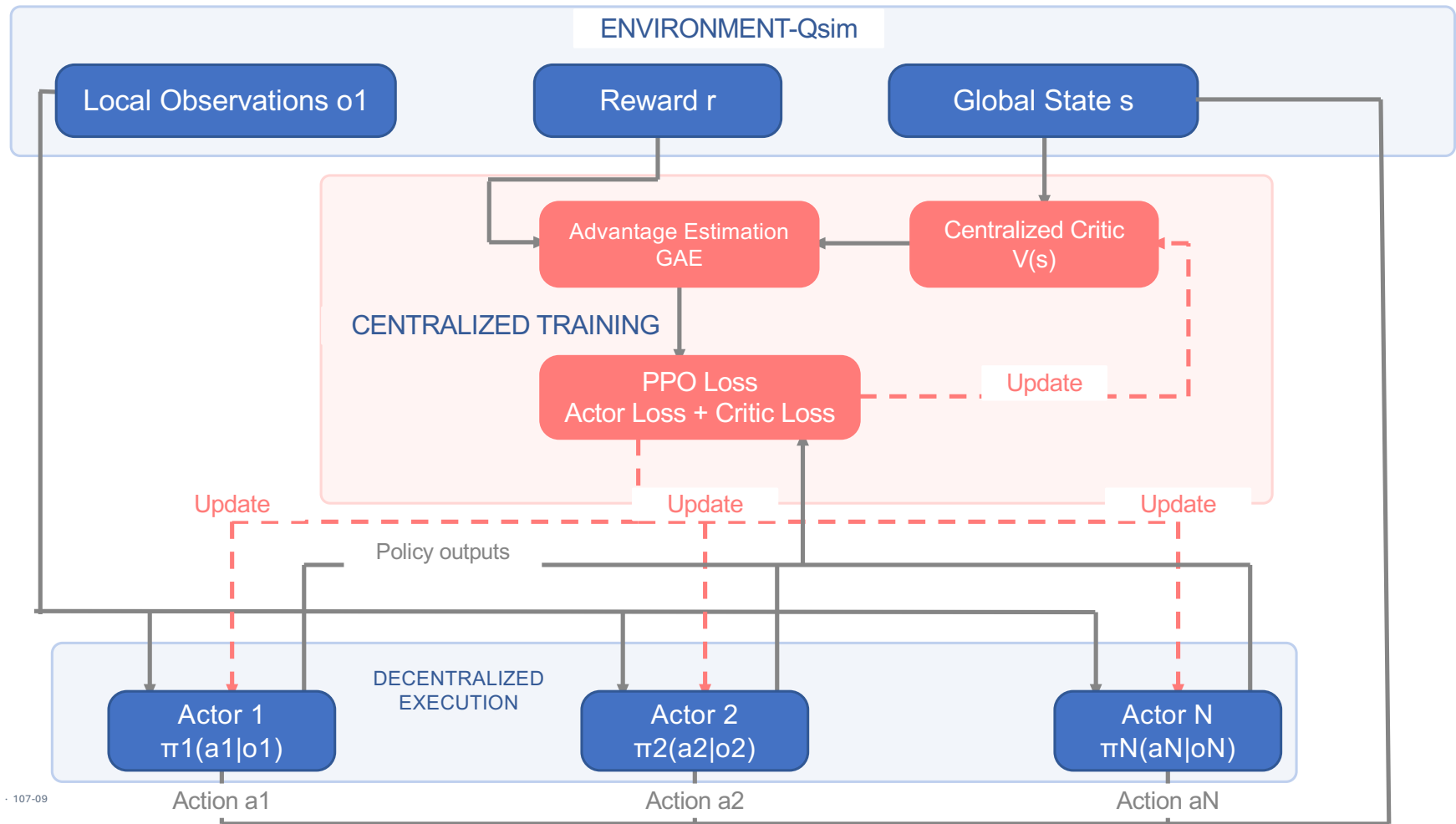
Mappo procedure – training



Mappo procedure – mappo takes over and perform decisions

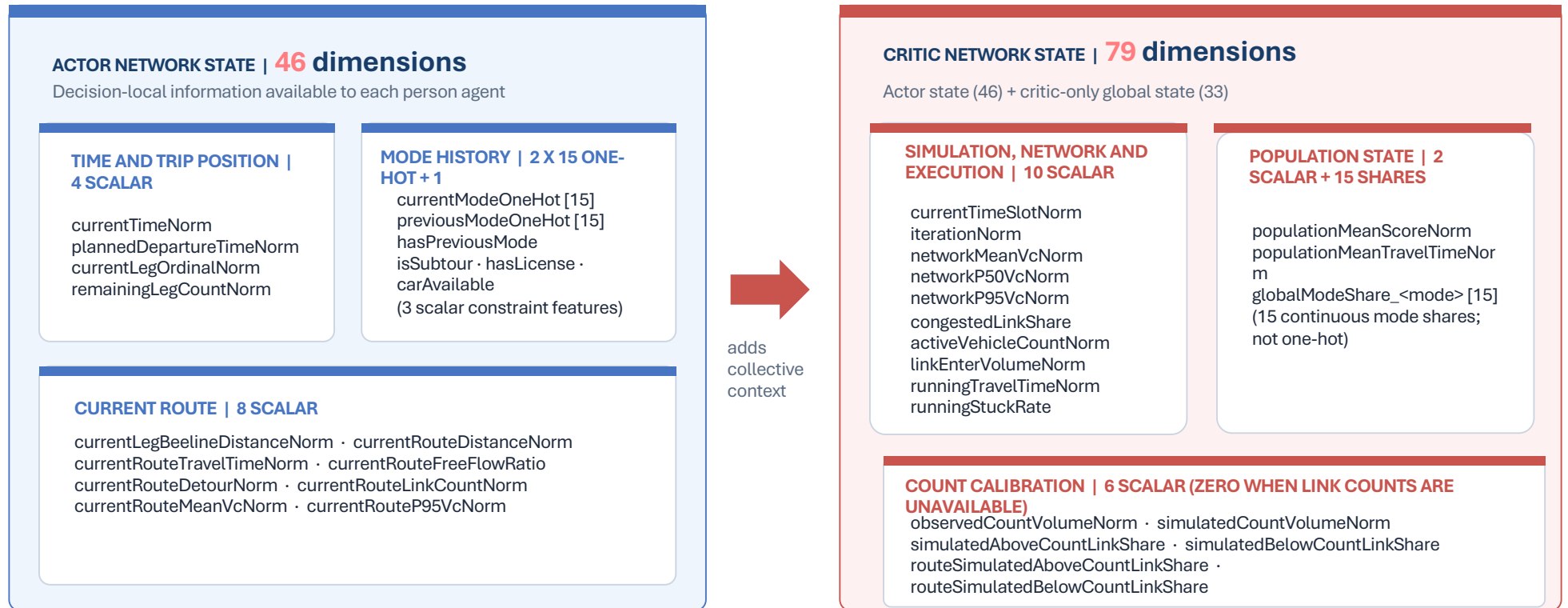


Mappo framework- typical actor-critic networks structure



MAPPO definition: actor and critic state dimension and rewards

All features are normalized or one-hot encoded. Interaction activities and legs are excluded when deriving decision-trip state.



MAPPO definition: reward functions

$$r(i,t) = 0.5 \Delta S_O(i,t) + \Delta S_L(i,t) + 0.5 \Delta S_D(i,t) \\ - \rho(i,t) - 150 \cdot z(i,t) \cdot \rho_G$$

MATSim score improvement

$$\Delta S_x(i,t) = S_x(i,t) - S^0_x(i,t), \quad x \in \{O, L, D\}$$

O = origin activity; L = decision trip; D = destination activity.

Each score is compared with the same person and trip ordinal in the baseline.

Local travel-time change

$$\rho(i,t) = [T(i,t) - T^0(i,t)] / T^0(i,t)$$

T is realized decision-trip travel time, measured in seconds.

Set $\rho(i,t) = 0$ when baseline travel time $T^0(i,t) \leq 0$.

Global travel-time change

$$\rho_G = (TT - TT^0) / TT^0$$

TT is the sum of valid decision-trip travel times in the current iteration.

Set $\rho_G = 0$ when TT^0 is invalid or non-positive.

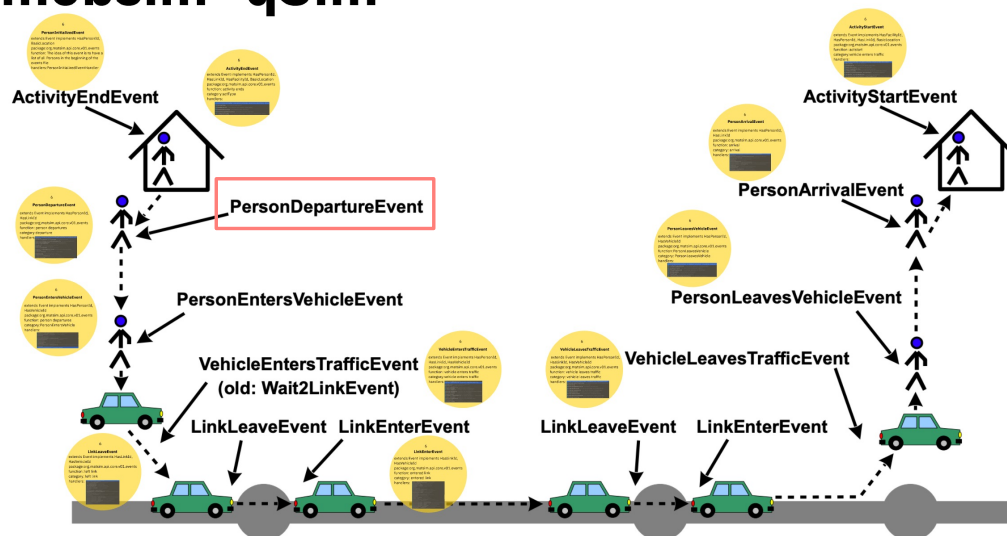
$z(i,t) = 1$ only at each person's final decision; otherwise, $z(i,t) = 0$.

i = person; t = decision index. Superscript 0 denotes the fixed baseline captured at MAPPO takeover.

Higher scores or shorter travel times increase reward. The final reward is not normalized or clipped.

Person action trigger point & action space

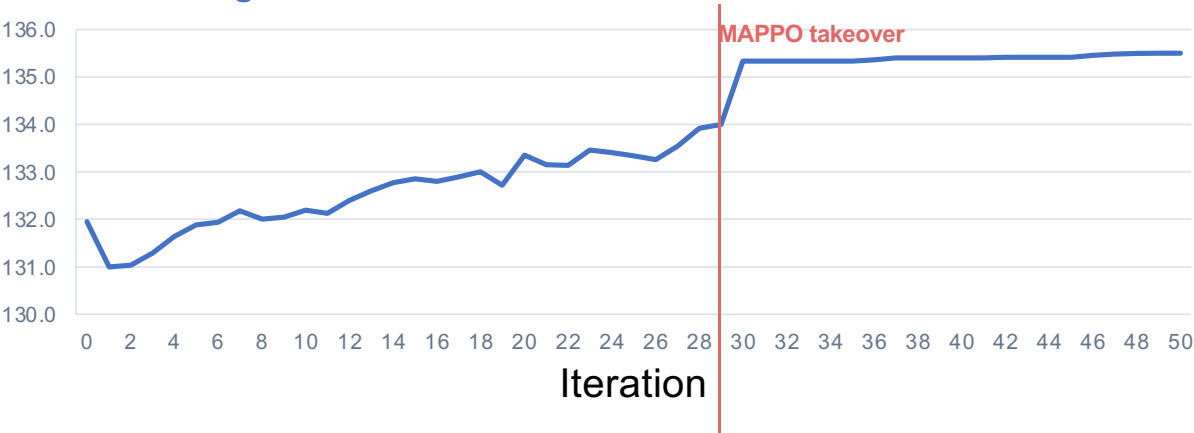
mobsim -qSim



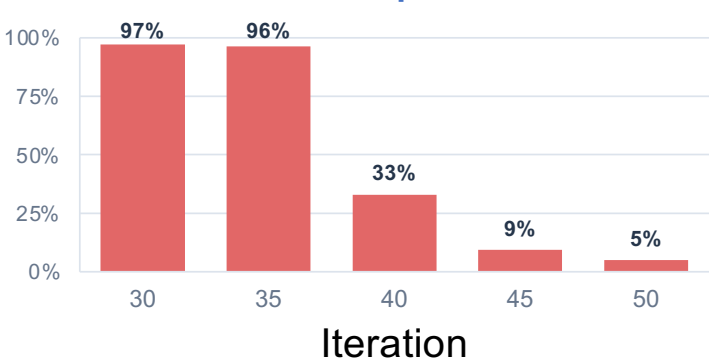
MATSim-MAPPO Run Evaluation

The run completed successfully, while the learned policy shifted toward car and lost acceptance over time.

MATSim average executed score



MAPPO candidate acceptance



MATSim outcome

134.00 to 135.50 **-17.1% to +9.6%** **87.9% to 98.6%**

Average score
+1.12% vs iter. 29

Global travel time
iter. 30 to iter. 50

Car mode share
iter. 29 to iter. 50

Model evaluation

+10.91 to -7.83 **0.00 to -19.21** **21.7 to 2,421.9**

Mean immediate
reward
iter. 30 to iter. 50

5th percentile reward
iter. 30 to iter. 50

Critic loss at epoch 20
iter. 30 to iter. 50

Assessment: integration and checkpointing passed. Performance remains unvalidated due to mode collapse, rising travel time, negative rewards and falling plan acceptance.

MAPPO Plan Selection: Safeguard vs Full Retention

Post-takeover selection rule

SELECTION SAFEGUARD

Evaluate MAPPO and existing alternatives in QSim, then retain the highest-score plan.

27.1%

4,872 MAPPO plans selected out of 18,000 candidates

Selection preserves a tested fallback when a MAPPO candidate underperforms.

FULL MAPPO RETENTION

After takeover, select the MAPPO candidate directly for every person and every iteration.

100%

MAPPO plan retention from iteration 31 to 50

This isolates the effect of removing the score-based selection safeguard.

Iteration 50 outcomes

Score safeguard

137.89

Average executed score

1,371 s

Average decision-trip travel time

Full MAPPO

134.69

Average executed score

1,321 s

Average decision-trip travel time

Selected-plan modal split

Score safeguard

47.8%

36.1%

16.1%

Full MAPPO

53.7%

20.8%

25.4%



Car



PT



Walk

In this controlled test, direct MAPPO retention lowers average decision-trip travel time by 3.7% relative to the safeguard, but reduces average executed score by 3.20 utility and shifts 15.3 percentage points away from PT. The score-based safeguard remains useful while the policy is still learning.

Scenario: GZ PT network · MAPPO retraining + on-policy

Server snapshot: 2026-09-03

Scenario scale

Population

986,424 persons

hpPop100w.xml.gz
roughly one million daily agents

Road network

80,982 nodes

150,928 links
gzInpoly_Simp_remod_pt_240129.xml

Public transport

1,879 lines

3,321 transit routes
145,174 transit vehicles

Run envelope

100 iterations

lastIteration = 100
MAPPO takeover starts after it.60

MAPPO and QSim configuration

Training regime and rollout data

runMode = retraining
onPolicy = true
collectTrainingData = true

it.88 training transitions: 8.15 GB
it.84–it.88 transition files: 38 GB
Full mappo-rollouts archive: 354 GB

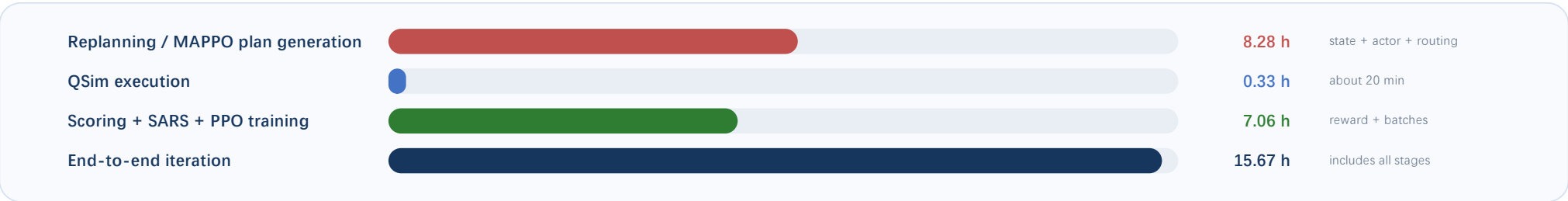
Simulation envelope

QSim end time: 24:30
flowCapacityFactor = 0.8217
storageCapacityFactor = 0.8217

networkModes = car
Counts input: linkCounts240407.xml

Observed runtime after MAPPO takeover (completed it.84–it.88)

Mean controller-stage wall-clock time; the bars are not isolated neural-network kernel timings.

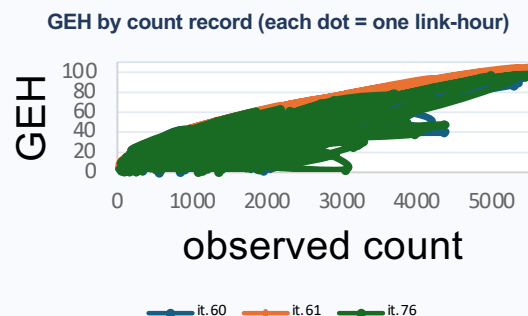
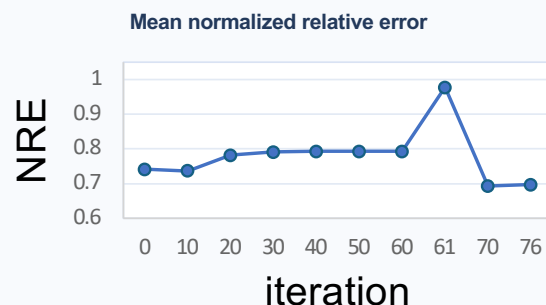


Interpretation: the measured “prediction” stage is the full replanning/MAPPO plan-decoding stage, while the measured “training” stage includes scoring, SARS construction, and PPO/BC updates. Training data are materialized as JSONL SARS transitions; dedicated intra-network timers have not yet been instrumented.

Guangzhou Scenario: Link Count Fit, GEH Record-Level Scatter, and PT Departure Deviations

Link count fit: a takeover disturbance followed by recovery to lower error

Red dashed line marks it.61, the MAPPO takeover



Representative iteration

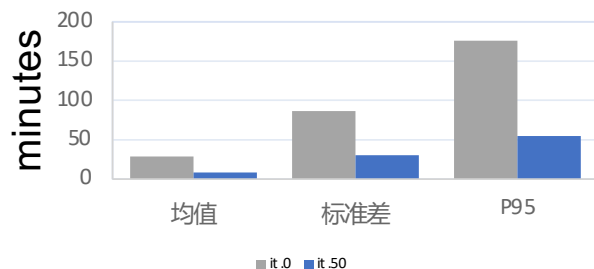
snapshots

NRE / mean GEH / GEH<5

iteration	NRE	GEH	GEH<5
it.0	0.741	28.86	6.1%
it.60	0.794	31.14	3.1%
it.61	0.978	41.24	0.0%
it.70	0.693	26.45	5.9%
it.76	0.697	26.78	6.3%

The it.61 cloud shifts upward across most observed-count ranges; by it.76 it shifts downward.

Observed PT departures versus timetable: marked pre-takeover convergence



Departure events

3.22m → 3.29m

it.0 → it.50

Departures delayed > 60 s

32.9% → 23.9%

Decrease by 9.0 pp

Median

0 s → 0 s

Many departures on time

Correct interpretation: observed schedule stability, not schedule-feedback evidence

• config-mappo-100iter-server6.xml does not set useExperiencedTransitSchedule; the default is false.

• Therefore, delay measures QSim departure deviation from the fixed timetable. The improvement from it.0 to it.50 reflects network and default-replanning convergence.

• MAPPO takes over at it.61. The retained count results through it.76 recover after the initial disturbance, but complete post-takeover event files are unavailable for an equivalent PT-departure comparison.

Next step: enable useExperiencedTransitSchedule and write events at shorter intervals to distinguish observed delays from a timetable-feedback effect.

Summary - aspects to be improved/done...

- dimensionless states with PT time table & volume count calibration
- Raise dimension like nerf svm might get more latent
- apply multi-agent/ heterogenous agents to the extension
- activity-trip-chain GT for auto parameters calibration, CMA-ES framework
- can it integrate with world models?



Milu

广东 深圳

convel@163.com

